MATTHEW PORCELLI

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PROFESSIONAL SUMMARY

Senior Network Engineer with 20+ years in telecommunications, including 17+ in service-provider network engineering at Cox Communications, specializing in IP/MPLS, DWDM/optical transport, QoS design, and network automation. Sets national engineering standards for the Cox Business Network, leads High-Level Design (HLD) for Nokia optical and IP/MPLS deployments, and owns capacity planning and lab/lifecycle budgeting. Builds network automation and lab tooling (Containerlab, Docker, Python, gNMIc, Prometheus/Grafana, MCP servers) adopted across engineering teams. Hands-on with Nokia 7x50/SR, Juniper MX, Cisco ASR/CRS, and Cisco Meraki platforms; BGP, IS-IS, OSPF, Segment Routing (SR-MPLS), and Carrier Ethernet (MEF).

CORE COMPETENCIES

Network Design & Planning: IP/MPLS Designs • DWDM/Optical Designs • QoS Strategy • Capacity Planning • Requirements-Based Hardware Selection • Production & Lab Budgetary Planning • Lifecycle Management

Automation & Platforms: Network Automation • Cisco Meraki • CI/CD Pipelines • REST APIs • MCP Servers • Tooling Development • Containerlab • Docker • Python • Linux

Testing & Observability: Prometheus • Grafana • Observability • Test Automation • Robot Framework • Regression Testing • Validation & Rollout Testing • Pre/Post-Check Automation

Governance & Documentation: National Market Governance • Engineering Standards • Technical Documentation • Methods of Procedure (MOPs) • Stakeholder Communication • Change Engineering & Validation

PROFESSIONAL EXPERIENCE

Cox Communications (Cox Business) — 2009 – Present

Sr Communications / Network Engineer III — National Engineering Design — 2020 – Present

- **Optical Design Lead:** Nationally lead Nokia Optical (1830 PSS) design across 3 regions and 11 markets; lead Access design (600+ nodes, 1,200+ services).
- **IP/MPLS Design Standards:** Set the design and configuration standards for Core (700+ routers) and Access (79,000+ routers) services across a network of 1.2M+ services; Nokia IXR-e2/IXR-e2c design and testing.
- **Meraki SD-WAN Underlay:** Contributed to underlay design for Cox Business SD-WAN; worked to understand overlay requirements and class of service; lab configuration and testing, including Meraki high-availability failover.
- **NOCaaS PoC:** Tested Meraki gear for NOCaaS alarm monitoring; testing showed it lacked required capabilities.
- **Evaluation & Validation:** HLDs and lab/regression-tested hardware evaluations, including EVPN over SR-MPLS/SR-TE.
- **Budget, Capacity & Lifecycle Planning:** Optical lifecycle budgeting and burden calculations; lab budgeting; Containerlab virtualization cost savings.
- **Automation & Tooling:** Containerlab/Docker lab on RHEL with Jenkins, gNMIc streaming telemetry, Prometheus, and Grafana; built Harbor, CLAB Orchestrator, and MCP servers (WSNOC, NFM-P/NSP, Elastic, Snowflake); ZTP Orchestrator proof of concept (DHCP Option 67 on a containerized Kea DHCP server); Endpoint (Nokia WSNOC lifecycle and budget tracking).

Sr Communications / Network Engineer III — National Sustaining Engineering — 2018 – 2020

- **QoS Standards:** Owned QoS policy standards (policers and shapers) for the network; QoS Standards Guide.
- **Tier IV Escalation:** Highest-tier escalation support; resolved Sprint (45+ circuits) and T-Mobile (85+ circuits) failed-SLA issues, plus IPVPN MTU failures, multicast QoS above 2 Gbps and 7250 IXR frame discards; Technical Bulletins and QoS Decision Tree adopted in 3 regions, 11 markets.
- **Video Products:** Engineered the underlay for Pro Idiom (multicast) and Contour (unicast IPTV); developed and lab-tested CPE-less design standards, co-authored national playbooks with Atlanta Video/Metro Engineering and OSS; led training across 11 markets.
- **RSAAT Test Automation:** Playbook scenarios for Spirent connectors and a 100G Orchestrator (100 simultaneous tests); helped retire end-of-support probes and standardized 100+ across markets; remote packet-capture MOPs.
- **Standards & Collaboration:** Authored and governed engineering standards and playbooks; led biweekly sustaining and monthly FSA/SA calls across 3 regions, 11 markets.

Network Operations Engineer II — Southwest Region **2009 – 2018**

- **Architecture & Planning:** Capacity plans and baseline configs for Nokia Core MPLS (75+ routers) and access rings (4,000+ devices).
- **Compliance & Deployment Standards:** Granite/Telcordia-compliant changes and remediation packages for Operations; deployment guides, with the DS3 circuit emulation guide (DACS integration) adopted nationally as a path off SONET.
- **RFP Design:** Lead designer for the ASU and Raiders Stadium RFPs; defined how the network would be built and leveraged for customer needs.
- **Cisco DWDM:** Capacity planning for 30+ nodes across 5 Southwest networks; designed a 96-channel Flex Spectrum network (Phoenix, 2017) and trained engineers on it.
- **Routing, Voice & TDM:** BGP, IS-IS, OSPF, MPLS and MEF circuit types (EPL, EVPL, EP-LAN, EVP-LAN) across Cisco, Juniper and Nokia/Alcatel-Lucent platforms; H.323/SIP voice and SONET/DWDM transport.

CERTIFICATIONS

Nokia NRS I (2016) • Cisco CCNP Route & Switch, SWITCH exam (2015) • Cisco CCNA (2005, recertified 2012)

EDUCATION

Valencia Community College: A.S., Computer Engineering Technology (Cisco Specialization), 2004; A.A., 2002

Cisco Networking Academy, 2004